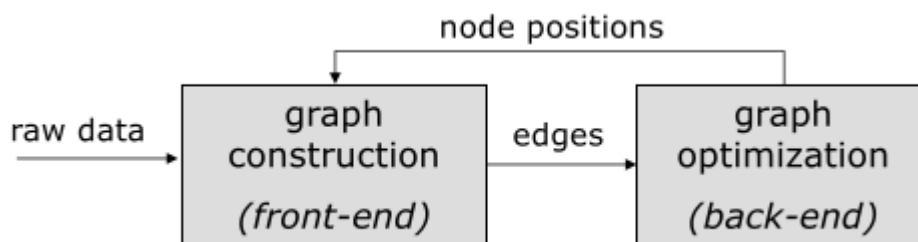


L17: Hierarchical Slam

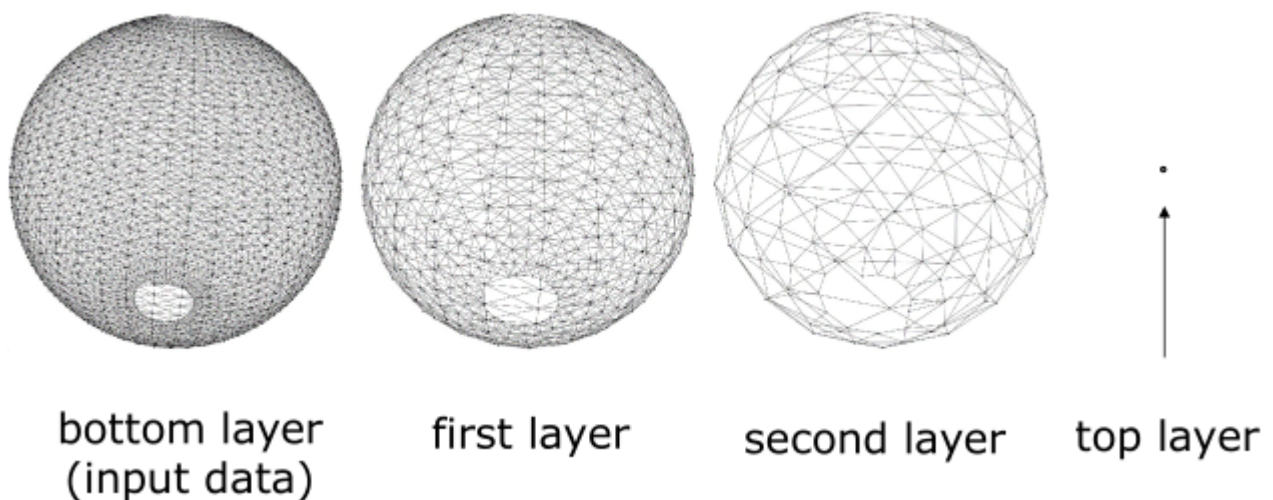
SLAM

- **Front-end:** Data association
 - Extracts constraints from the sensor data
- **Back-end:**
 - Optimises pose-graph to reduce introduced error by constraints



- Intermediate Solutions are needed to make good data associations

Hierarchical SLAM



- No need to optimise the whole graph, whenever a new observation is obtained

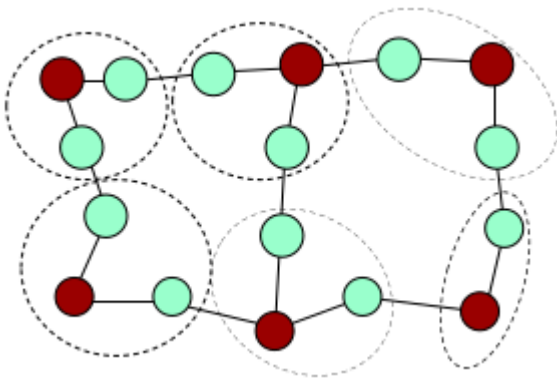
Motivation

- SLAM front-end seeks for loop-closures
- Requires to compare observations to all previously obtained ones
- In practice, limit search to areas in which the robot is likely to be
 - **In which parts of the graph to search for data associations**
- **Hierarchical Approach:**
 - **Insight:** Loop closing points, one does not need the perfect global map

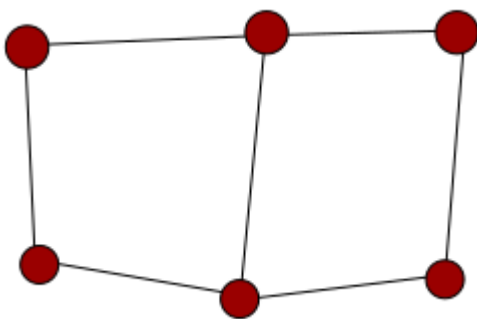
- **Idea:** Correct only core structure of scene, not overall graph
- Hierarchical pose-graph as a sparse approximation of original problem
- Exploits SLAM facts:
 - Robots moves successivly: No teleportation to location
 - Sensors have a limited range

Key Idea of the Hierarchy

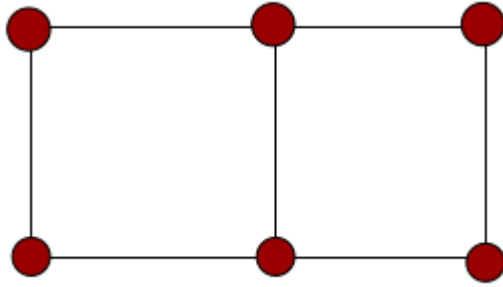
- Input is the dense graph
- Group nodes of the graph based on their local connectivity
- For each group, select one node as a "representative"



-
- Representatives are the nodes in a new sparsified graph (upper level)
- Edges of the sparse graph are determined by the connectivity of the groups of nodes
- Parameters of the sparse edges are estimated via local optimisation
- **Process is repeated recursively**

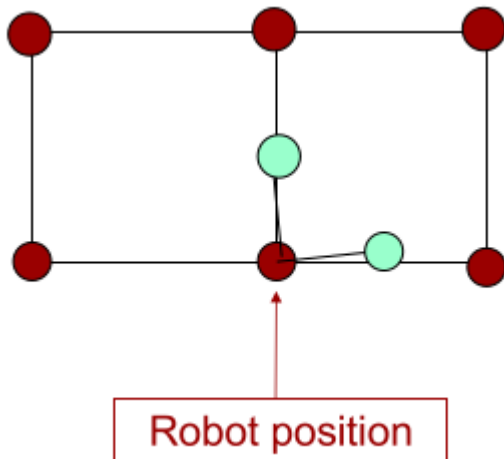


-
- Only the upper level of the hierarchy is optimised completely



■

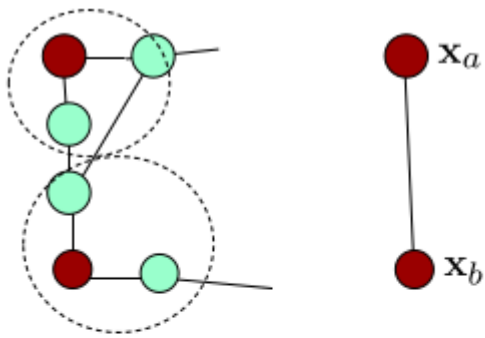
- Changes are propagated to the bottom levels - only close to the current robot position
- Only this part of the graph is relevant for finding constraints



■

Construction of Hierarchy

- **Group Construction**
 - Simple distance-based decision
 - Representative: **First node** of a new group
- **Time** of downwards information propagation
 - Only when $\exists$ inconsistencies
- Construction of edge in sparsified graph:
 - Determination of edge parameters:
 - Given 2 connected groups
 - Compute virtual observation Z and information matrix Λ for new edge:
 - Optimise 2 sub-groups independently from the rest
 - Observation is the relative transformation between the 2 representatives
 - Information matrix is computed from the diagonal block of the matrix



Inverse of the $[b,b]$
block of $\mathbf{H}^{-1}$

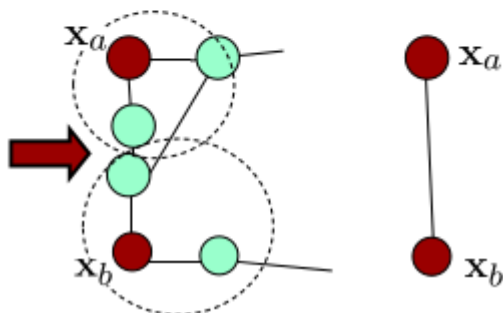
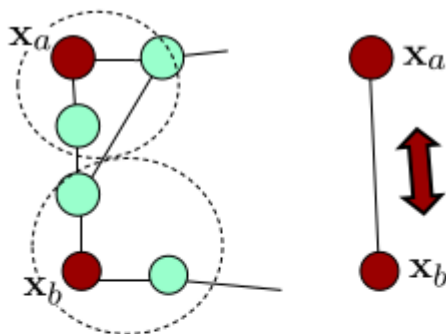


$$\Lambda_{ab} = (\mathbf{H}_{[b,b]}^{-1})^{-1}$$

■

- Downwards Information **Propagation strategy**

- All representatives are nodes from the lower (bottom) level
- Information is propagated downwards by transforming the group at the lower level
 - Using a rigid body transformation



◦

- Only if lower level becomes inconsistent, optimize the lower level

Best Possible Map

- Run optimisation on lowest level
 - (at end)

- For offline processing: With all constraints
 - Hierarchy helps convergence faster in case of large errors
 - One Pass up (construct edges) - One Pass down
 - sufficient

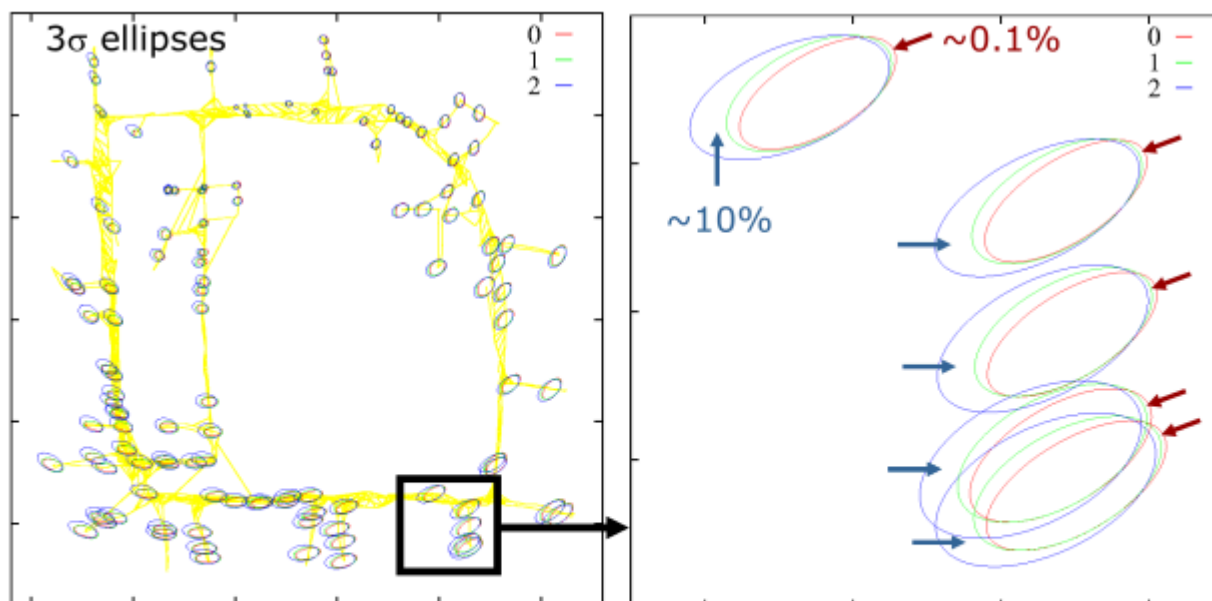
Consistency

- Top level hierarchy compared to original input
- Probability mass of marginal distribution in
 - Highest level vs.
 - True Estimate
 - original problem, lowest level

	Prob. mass not covered	Prob. mass outside
Intel	0.10%	10.18%
W-10000	2.53%	24.05%
Stanford	0.01%	7.88%
Sphere	2.75%	10.21%

low risk of becoming overly confident

one does not ignore too much information



Red: overly confident ($\sim 0.1\%$ prob. mass)

Blue: under confident ($\sim 10\%$ prob. mass)

Conclusions

- Hierarchical pose-graphs:
 - Provide approximate solution fast
 - Support efficient data associations
 - Designed for online mapping (interplay of optimisation and data association)
 - Higher levels can be seen as simplified SLAM problems